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AI-Powered Traffic Volume Prediction

This project is specialized in Smart City solutions.
Our mission is to help governments and businesses predict traffic congestion before it happens using Machine Learning

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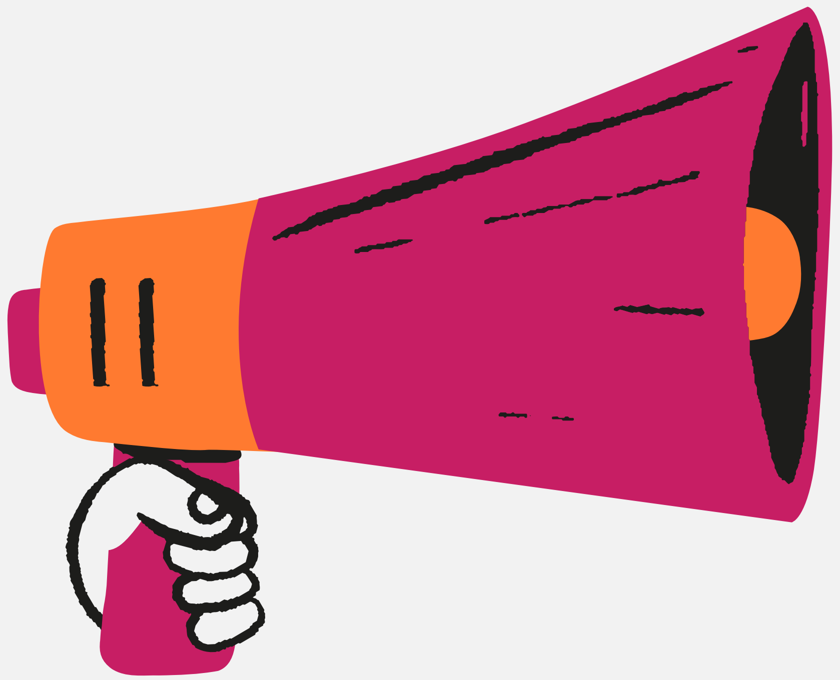
Team Members:

1.Mariet Ramzy Latif
2.Nervana Emad Waheeb
3.Habiba Ahmed Mohammed
4.Aryam Awad Mahmoud
5.Mena Jacop Wahba

Business Problem | Current Challenges



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Fuel Waste

We align content goals with overall business objectives and marketing strategy

Traffic Congestion

Heavy traffic causes long travel times, increases driver frustration, and reduces the overall efficiency of road transportation.

Increased Pollution

Traffic congestion increases carbon dioxide (CO_2) and other harmful emissions, contributing to air pollution and climate change.

Poor Traffic Planning

Without accurate traffic predictions, city planners struggle to optimize road infrastructure, traffic signals, and transportation policies.

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Our Solution: AI Traffic Prediction System

We developed an AI-powered Traffic Prediction System that leverages historical traffic and weather data to forecast future traffic volume. By analyzing patterns and trends, our solution enables proactive traffic management instead of reactive decision-making.

01. Accurate Traffic Forecasting:

Predicts traffic volume for future time periods using machine learning models.

02. Data-Driven Decision Making:

Provides reliable insights to support traffic authorities and city planners.

03. Scalable Solution:

Can be adapted to different cities and integrated with various traffic monitoring systems.

04. Smart City Support:

Contributes to intelligent transportation systems by improving road efficiency and reducing congestion.

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Data Overview:

Dataset Source:

- Kaggle

Weather indicators:

- pollution index
- humidity
- wind speed/direction
- visibility
- dewpoint
- temperature
- rain/snow
- clouds
- weather type

Time indicators:

- hour,
- day of week
- month
- year
- holidays
- weekend

Target variable: traffic volume (vehicles/hour)

Side stats: dataset starts 2012, 15 engineered features, 100% cleaned data

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Process Flow:

EDA

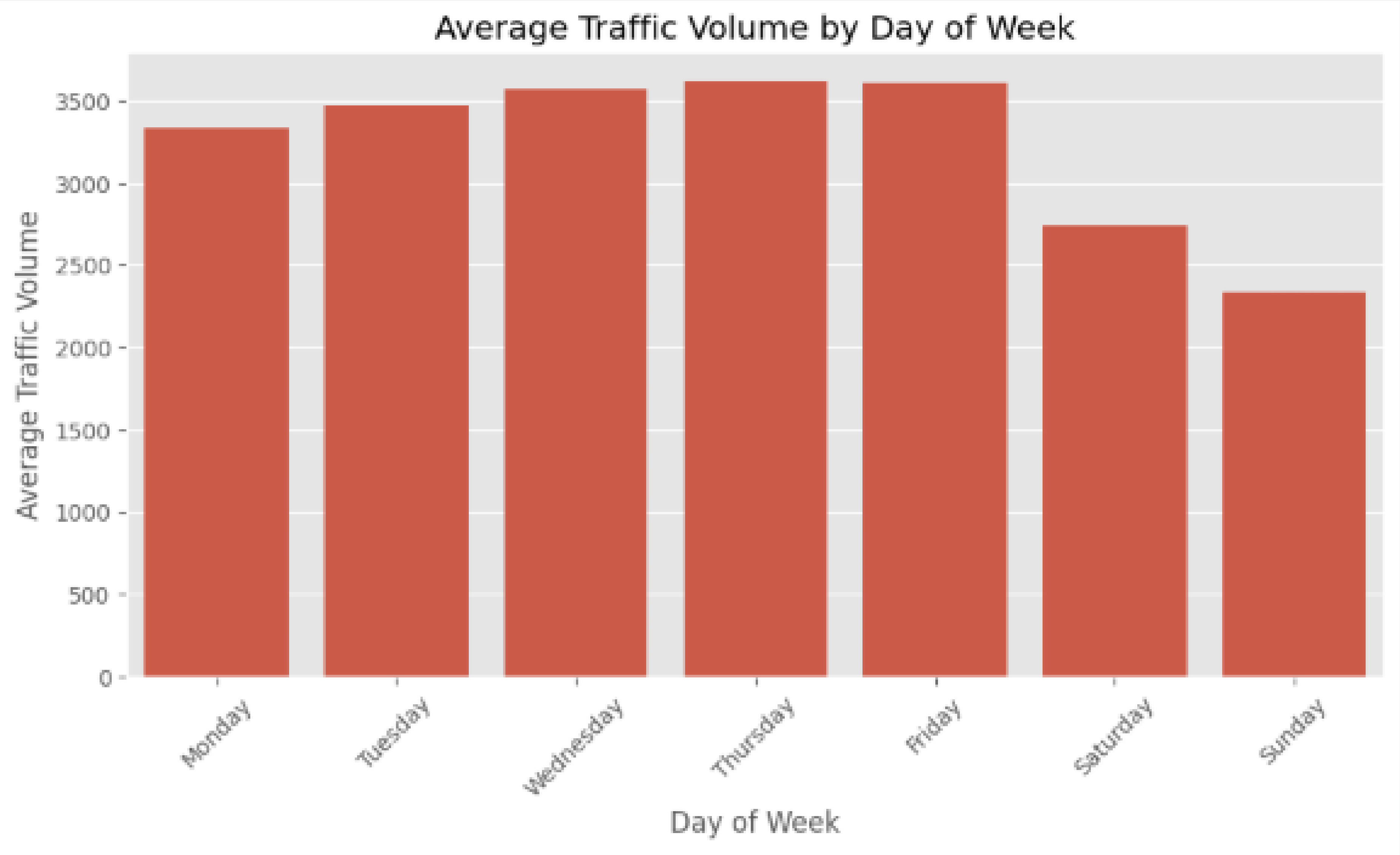
Feature Engineering

Model Building

Hyperparameter Tuning

Deployment

“What the data tells us before modeling”



Time patterns: bar chart of average traffic volume by day of week.

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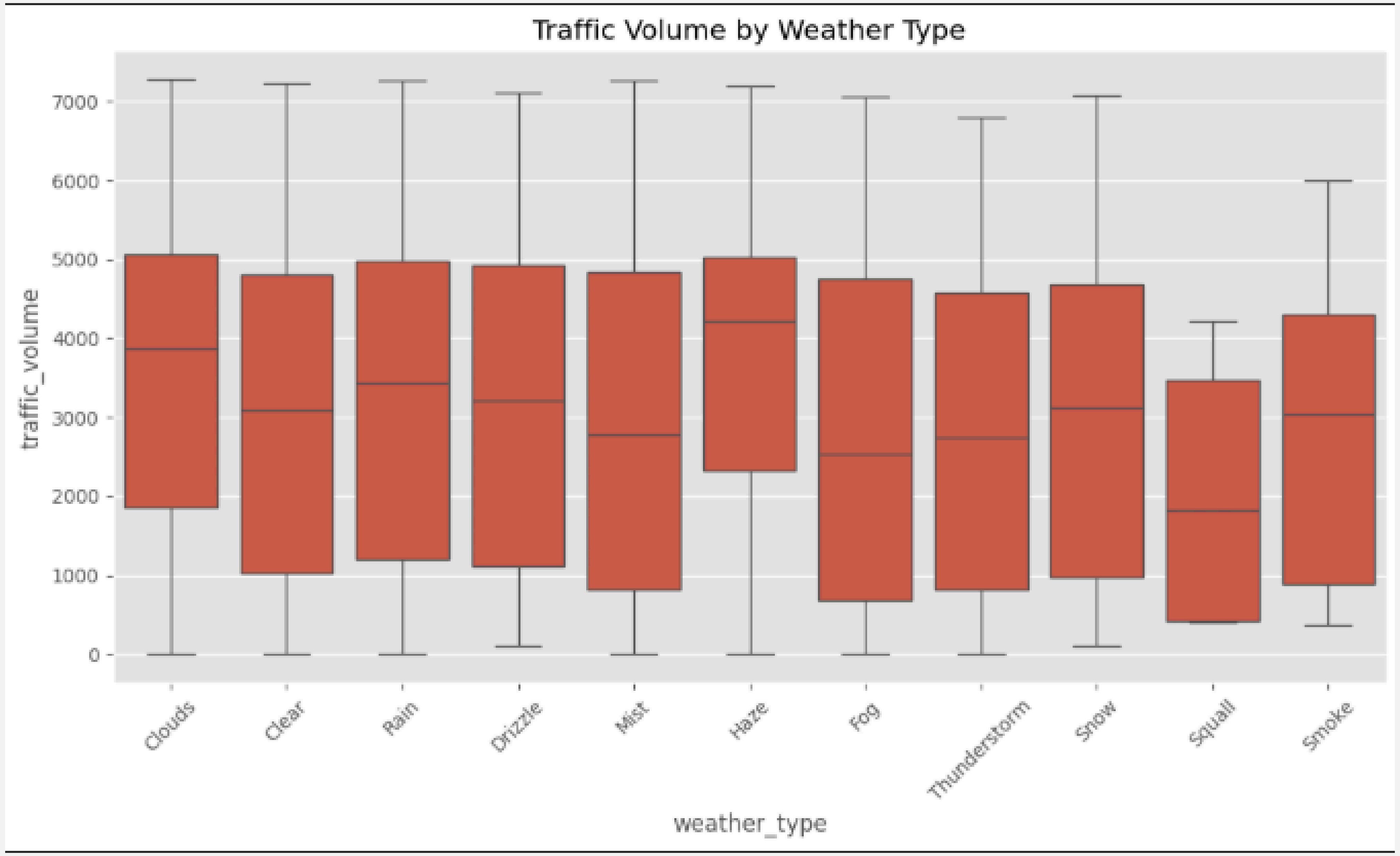
“What the data tells us before modeling”



Weather impact: box plot or bar chart of traffic volume by weather_type (Clear/Rain/Snow/etc.)

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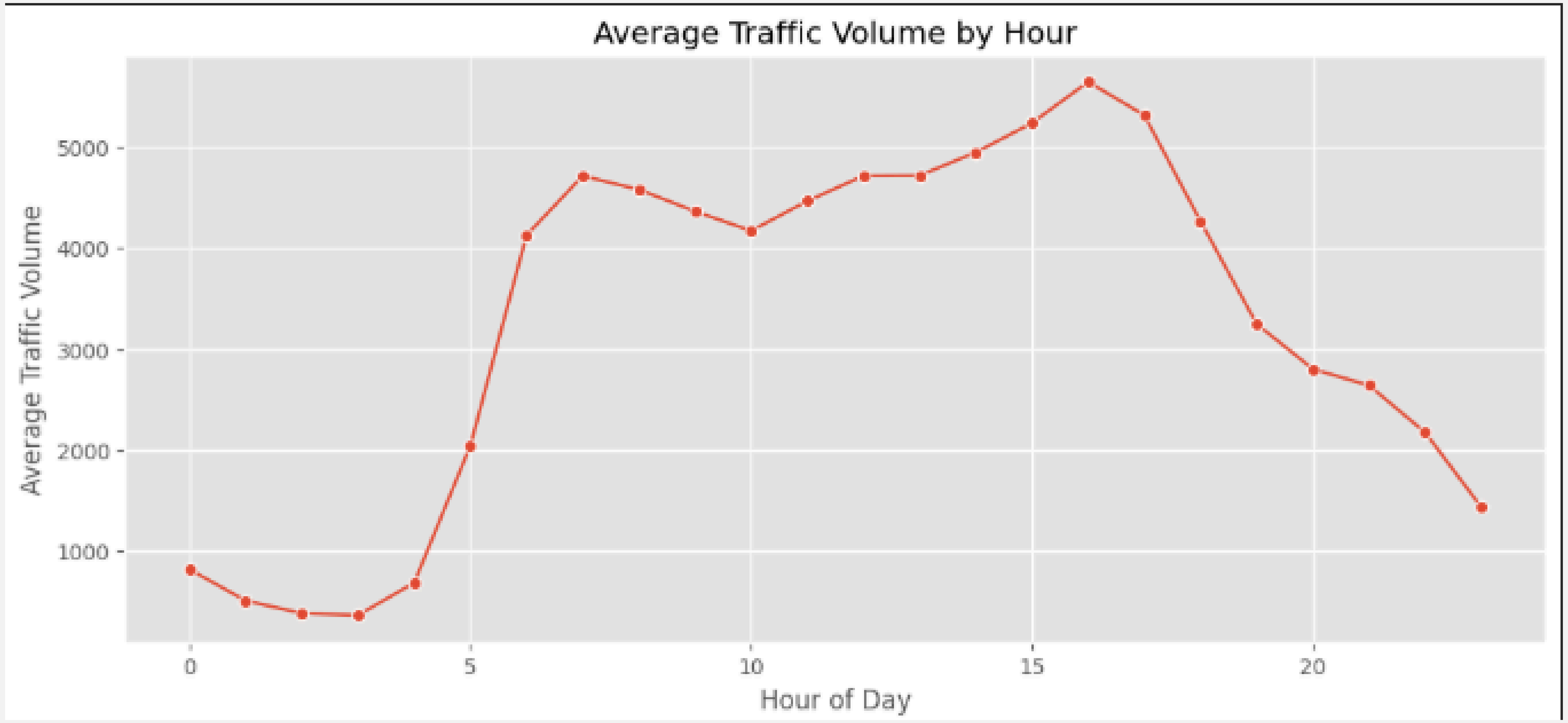
“What the data tells us before modeling”

Time patterns: line/bar chart of average traffic volume by hour of day.



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Feature Engineering Highlights:

Cyclic hour encoding

(sin/cos) — so hour 23 and hour 0 stay "close"

Rush hour flags

(7–9 AM, 4–7 PM)

Weekend/holiday flags

Date decomposition

(day, month, year, day-of-week)

Model Comparison:

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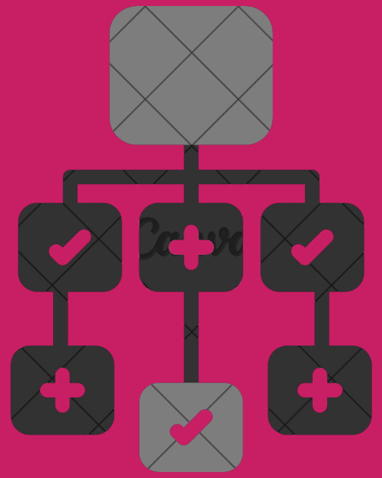
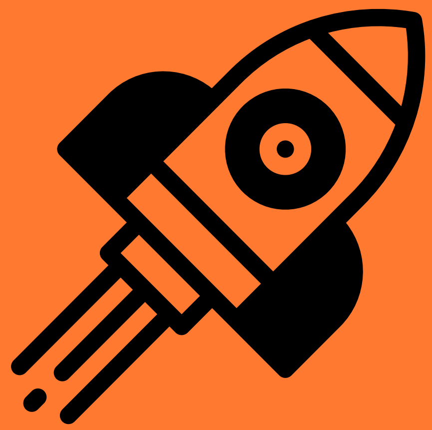
Linear Regression 75%



Random Forest 93%



XGBoost 94%

XGBoost highlighted as best

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Deployment:

Mock app screenshot/description:

streamlit interface (date/time, weather inputs, predict button)

Three info cards:

Streamlit (interactive UI), Cloudflare Tunnel (public secure access), Saved Pipeline (consistent preprocessing)

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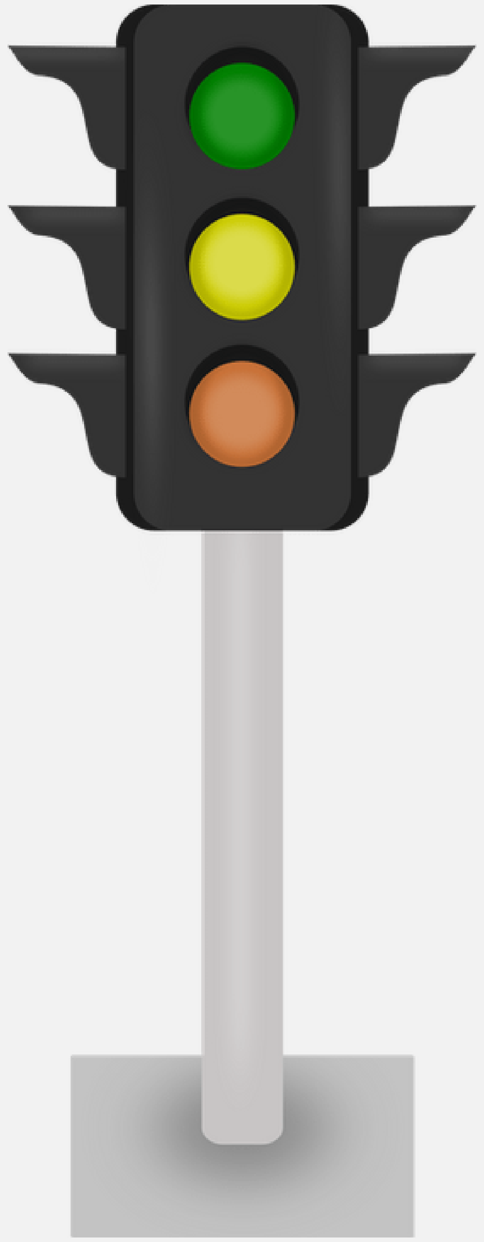
Roadmap :



01.

Phase 1:

Connect to live weather API

02.

Phase 2:

Expand to multiple locations/cities

03.

Phase 3:

Build REST API + mobile app for partner integration

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Thank You